

Abstract

Pacific herring (*Clupea pallasii*) spawning creates a concentrated coastal resource pulse, but broad-scale bird associations are difficult to estimate from opportunistic observations. We linked eligible submitted complete eBird checklists to event-level Fisheries and Oceans Canada spawn records in registered spatial and temporal windows. Mixed models estimated checklist detection and positive numeric count conditional on detection, with registered effort, observer, temporal, and spatial adjustment. The primary frames contained 217,200 Strait of Georgia and 8,584 West Coast Vancouver Island checklist events. Registered guild and priority-species coefficients varied by region, outcome, and taxon; Surf Scoter and Short-billed Gull coefficients were positive in both primary regions and both response components. Matched 2-km and dominant-observer sensitivities each preserved the matched-reference sign in 15 of 16 components, although the protected sensitivity set retained 43 singular-fit warnings. None of 64 matched whole-bundle placebo components had a Benjamini-Hochberg q -value below 0.05. The prespecified Strait of Georgia specificity panel was non-null for Gadwall (beta = 0.234, 95% CI 0.132–0.337; $q = 7.28 \times 10^{-6}$) and Northern Shoveler (beta = 0.628, 0.527–0.729; $q = 6.56 \times 10^{-34}$). The registered analyses provide confirmatory observational evidence of checklist-conditional associations between recorded local herring-spawn exposure and bird responses. Remaining observation-process limitations and the non-null specificity panel constrain causal and simple ecological-specificity interpretations. The results do not estimate population abundance, biomass, occupancy, migration, or individual movement.

Keywords: Pacific herring; eBird; community science; coastal birds; resource pulse; mixed models; falsification analysis